Mastering AI & Large Language Models: Architecture Guide

Transformer Foundations, Attention Mechanisms & Next-Token Inference

SkillForge AI Engineering Series - Certified Curriculum Materials

1. Transformer Foundations & Tokenization

Tokens are numerical representations of subwords processed by the vocabulary embedding matrix.

High-dimensional vector embeddings map semantic relationships into geometric latent space.

Positional encodings inject sequential order awareness into non-recurrent Transformer layers.

2. Multi-Head Scaled Dot-Product Self-Attention

Query Q, Key K, and Value V projections form dynamic contextual attention scores.

Formula: AttentionQ, K, V = softmaxQ * K^T / sqrtd_k * V.

Multi-head attention allows the model to attend to information at different subspace representations.

3. Production Deployment & Quantization (LoRA / QLoRA)

Freeze original weights W_0 and inject low-rank decomposition matrices A and B rank r << d.

QLoRA normalizes weights to 4-bit NormalFloat NF4 for 75% GPU memory reduction.

Continuous batching via vLLM PagedAttention achieves 4x-8x inference throughput.